CRYPTETHERA – The Trust Layer for India’s Power Grid

"We don't just meter power . We meter trust & integrity. "

TEAM - PASSIONATE PROBLEM SOLVER

- SAMEER SAAWAN

PROBLEMS

- India loses ₹1.2 lakh crore (~$14B) annually to power theft.
- Consumers face fake bills (₹3.8 lakh cases), suicides, and blackouts.
- Industrial investors avoid high-theft states → jobs & growth lost.
- Corruption: juniors coerced to fake records, systemic leakage.

Digital India
Power To Empower

RECOGNITION :
IRIS 2024 - SHORTLISTED AMONGST THE FINALISTS


ECD 210
Electronic Trivector Meter ER300P
12466494

THE PROBLEM

r/uttarpradesh • 1 yr. ago
[deleted]

India's Electricity Corruption: Why don't private player take over UPPCL

Ask UP

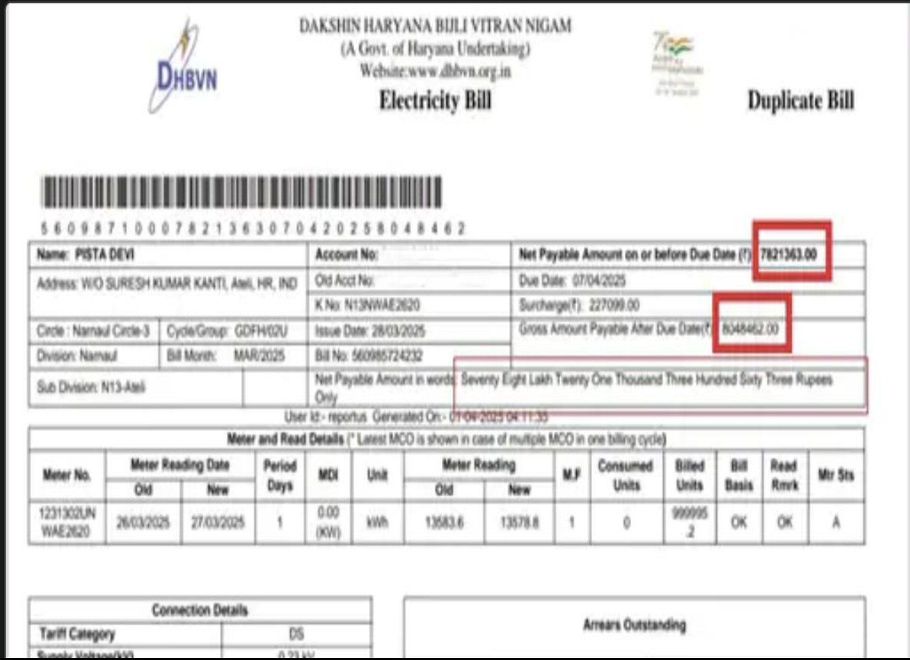
Applied for an electricity meter online on May 24th, thinking it would be a bribe-free process. Boy, was I wrong!

FARMER GETS 78 LAKHS BILL !

In Narnaul, Haryana, a farmer received a one-day electricity bill of over Rs 78 lakh. The bill was sent just after he upgraded his meter. The family was shocked to see the amount and is now running around trying to get the bill corrected. Officials say the bill was caused by a mistake in meter reading and will be fixed.

Rs 78 Lakh bill sent after load increase

Farmer Suresh Kumar from Kanti village had increased the electricity load at his home. His house, built on Nawadi land, earlier had a 2-kilowatt meter in the name of his wife, Pista Devi. He recently upgraded it to 3 kilowatts and had paid Rs 1,717 for the earlier connection.


DAKSHIN HARYANA BIJLI VITRAN NIGAM
(A Govt. of Haryana Undertaking)
Website:www.dhbvn.org.in
Electricity Bill

Duplicate Bill

5 6 0 9 8 7 1 0 0 0 7 8 2 1 3 6 3 0 7 0 4 2 0 2 5 8 0 4 8 4 6 2

Name: PISTA DEVI	Account No:	Net Payable Amount on or before Due Date (₹) 7821363.00
Address: W/O SURESH KUMAR KANTI, Ateli, HR, IND	Old Acct No:	Due Date: 07/04/2025
	K No: N13NWAE2620	Surcharge(₹): 227099.00
Circle : Narnaul Circle-3 / Cycle/Group: GDFH/02U	Issue Date: 28/03/2025	Gross Amount Payable After Due Date(₹) 8048462.00
Division: Narnaul / Bill Month: MAR/2025	Bill No: 560985724232	
Sub Division: N13-Ateli	Net Payable Amount in words: Seventy Eight Lakh Twenty One Thousand Three Hundred Sixty Three Rupees Only	

User Id:- reportus Generated On:- 01/04/2025 04:11:35

Meter and Read Details (* Latest MCO is shown in case of multiple MCO in one billing cycle)

Meter No.	Meter Reading Date Old	Meter Reading Date New	Period Days	MDI	Unit	Meter Reading Old	Meter Reading New	M.F	Consumed Units	Billed Units	Bill Basis	Read Rmrk	Mtr Sts
1231302UN WAE2620	26/03/2025	27/03/2025	1	0.00 (KW)	kWh	13583.6	13578.8	1	0	999995.2	OK	OK	A

Connection Details

Tariff Category	DS

Arrears Outstanding

PROBLEMS

Ground Reality – Voices of Consumers

- *“We got a ₹7.8 lakh bill, I almost committed suicide.” – Quora (2023)*
- *“Juniors are forced to manipulate billing records by seniors.” – Reddit (2022)*
- *“Power theft blackouts made us shut our factory.” – News Report, Bihar, 2021*

Hyderabad: Domestic consumer gets Rs 3.81 lakh electricity bill; officials admit mistake

Ch Sushil Rao / TNN / Updated:
Jun 12, 2018, 16:49 IST

SHARE AA FOLLOW US

HYDERABAD: A domestic consumer was given a shock by the electricity department which generated a bill for Rs 3,81,571. D Swarupa, a resident of Srinivasanagar, Boduppal in the city has been asked to pay the huge amount for a billing which was done for 32 days

THE NEGATIVE IMPACT ON SOCIETY

Poor Industrial Growth:
Investors avoid regions like Bihar due to unreliable power—killing job creation, stalling economic growth, and wasting immense industrial potential.

Damaged Appliances:
Voltage fluctuations from overloaded or tampered lines often burn out appliances, causing financial loss and fire hazards in homes.

Unfair Load Shedding:
Areas with legitimate consumption often face load shedding caused by theft elsewhere—penalizing honest consumers with forced outages.

Lack of Trust in Utilities:
When people see others steal electricity with impunity, they lose faith in governance and public systems—leading to further lawlessness

OPPORTUNITIES

- $60B+ Indian electricity distribution market, $100B+ global
- Govt. push: Smart Meter Mission 2025, Smart Cities, Digital India, Electricity subsidies during elections
- The gap: **Trust + Transparency + Predictive maintenance**
- **No existing fool proof tech**

THE SOLUTION

- **Cryptethera = Blockchain + IoT + AI**
- **IoT Meters** → tamper-proof, signed consumption data
- **Blockchain Backbone** → immutable billing, no manipulation
- **AI Intelligence** → real-time anomaly detection & predictive maintenance
- **Validated @ IRIS 2024 (96.7% accuracy)**
- **Machine Learning for Tamper Detection:**
- **Predictive Maintenance (LSTM):**
  Time-series models learn electrical patterns and detect future equipment failures—reducing blackouts and operational costs.

Blockchain + AI = Corruption & Theft Killer

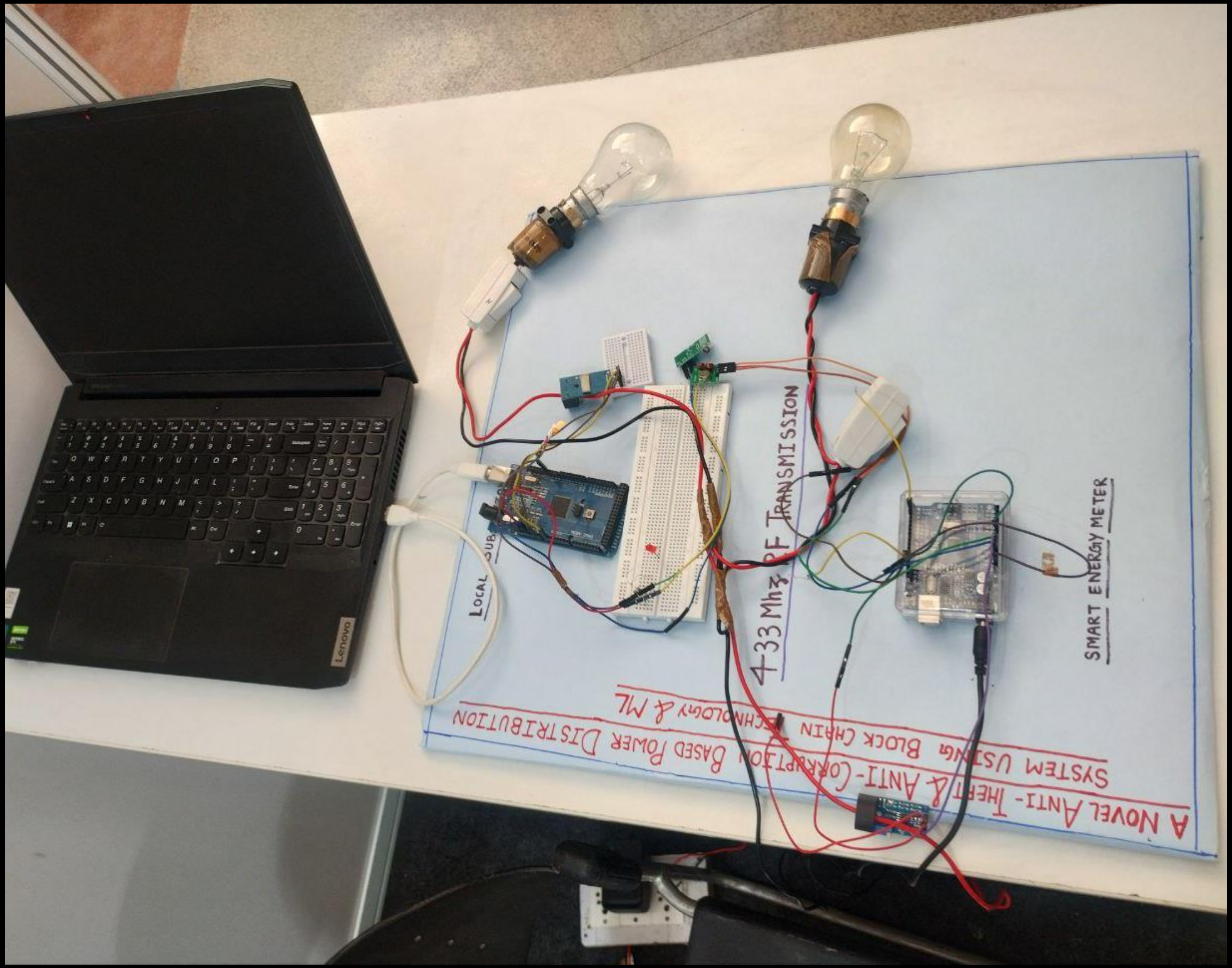
- Meters sign data → lower posts can't be forced to fake
- Smart contracts auto-generate bills → no manual tampering
- AI detects theft anomalies in real-time → logged immutably on blockchain
- Predictive maintenance alerts → blackouts & failures prevented
- Regulators & consumers see the same trusted truth

HOW IT WORKS

Meter → Gateway → IPFS + Blockchain → AI → Consumer & DISCOM Apps

- Meter signs data (secure chip)
- Gateway batches → Merkle root stored on blockchain
- AI flags theft & predicts failures → logged immutably
- Smart contracts auto-generate bills
- Consumer + DISCOM see *exact same truth*

POC AND MVP



MVP VERSION 01 (2024 : Depicted at IRIS NATIONAL FAIR

- **Prototype Validation – Tested Proof of Concept (POC)**
- ✅ 96.7% theft detection accuracy in real-world test cases
- ✅ AES-secured smart meter firmware successfully signed readings
- ✅ Blockchain ledger validated tamper-proof storage
- ✅ AI anomaly detection flagged theft in real-time
- ✅ Pilot-scale LoRaWAN + Wi-Fi tested in urban & rural conditions

CUSTOMER SEGMENT

Primary Users:

- State Electricity Boards (DISCOMs): Need real-time detection to reduce losses
- Urban Energy Cooperatives & Private Grids: Require scalable, tamper-proof metering
- Smart Meter Manufacturers: Can integrate our system for built-in fraud prevention
- Government Bodies: Can adopt it to ensure transparency in high-theft zones

📍 Ideal Regions (India Focus):

High-theft states: Uttar Pradesh, Bihar, Jharkhand, Madhya Pradesh, Odisha

Urban slums + rural grids = highest demand for decentralized protection

Already aligned with Govt. Smart Meter Mission 2025

USP

HOW OUR SOLUTION OUTPERFORMS EXISTING SOLUTIONS

Feature / Capability	Genus / Secure / L&T	Tata Power Grid	Our System (Cryptethera)
Real-time Theft Detection	❌ No	❌ Manual only	✅ ML-powered real-time detection
Tamper Resistance	⚠️ Basic Hardware	⚠️ Sensor Only	✅ AES ping + blockchain logging
Blockchain Immutability	❌ Not used	❌ Not used	✅ Used at all levels
Substation-Level Anomaly Detection	❌ None	❌ None	✅ Graph clustering + Isolation Forest
Unauth Device Blocking	❌ SIM-only whitelisting	❌	✅ Cryptographic device registration
Transparency & Auditability	⚠️ Partial (internal)	⚠️ Internal only	✅ Open audit logs, verifiable by DISCOM
Network Flexibility (RF/LoRa/WiFi)	❌ SIM-dependent	❌ SIM or Wired	✅ Modular, hybrid-ready
Built for Rural & Urban India	⚠️ Urban optimized	⚠️ Limited	✅ Optimized for tier-2/3 cities & villages

REAL WORLD IMPACT – ECONOMIC, SOCIAL AND STRATEGIC

1. Economic Impact

India loses ₹1.2 lakh crore (~$14B) annually to power theft → Source: Ministry of Power & EESL Reports

Cryptethera-like systems could reduce **up to 40% of AT&C (Aggregate Technical & Commercial) losses** in high-theft zones

Even 1% savings = **₹1,200+ crore** returned to DISCOMs annually

Consumers pay less as theft burden is lifted

Subsidy leakage prevented = better use of taxpayer funds

- **Social Impact**
- **Rural consumers often overcharged due to nearby theft**
  → *Vadodara case: ₹7.8 lakh bill for honest household*
- *Gorakhpur case (2022):* Smart meters bypassed → consumers blamed
- Political interference enables corruption - *Sambhal MP fined ₹1.91 Cr for theft*
- With Cryptethera, **truth is mathematically provable**
  → Honest users are protected; fraud is exposed, regardless of power
- ✅ *Empowers poor and middle-class users who currently pay for what others steal.*

Governance & Systemic Trust Impact

- Brings **blockchain-level auditability** to DISCOMs
- Reduces meter reader bribery, underreporting, and data tampering
- **Supports India's Smart Meter National Program** — aligned with:
  - Digital India
  - UDAY reforms
  - Revamped Distribution Sector Scheme (RDSS)
- ✅ *Enables transparent governance with live dashboards and tamper-proof logs.*

- **4. Long-Term National Impact**
  - Supports energy equity across cities, towns, and rural grids
  - Enables **climate goals** by ensuring reliable usage data
  - Encourages investor trust in Indian energy infra
  - Creates an open model replicable across other utilities (water, gas)
- ✅ *Scalable as a full “Trust Layer” for India’s digital infrastructure.*

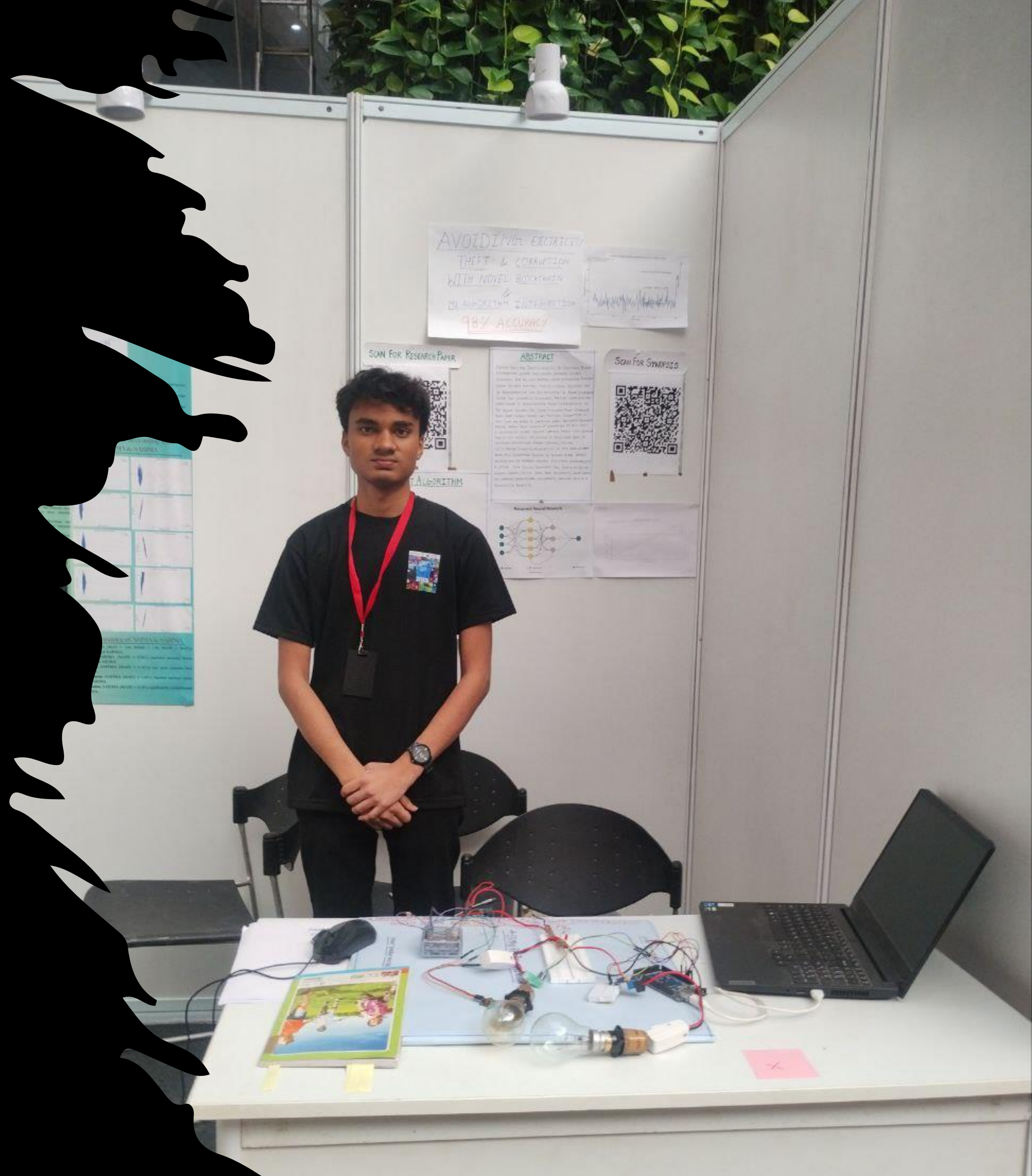
Revenue System

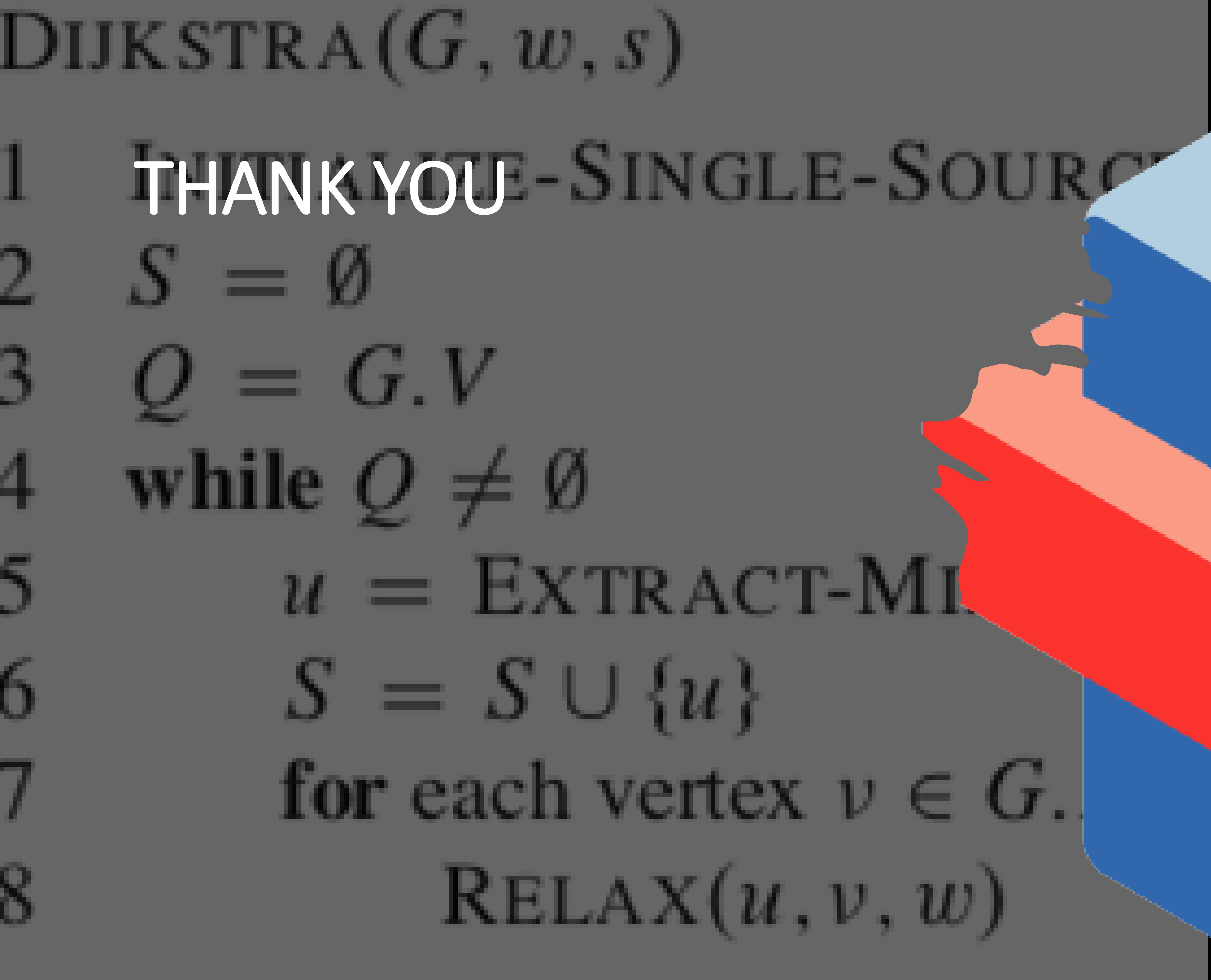
- 🔁 Per-Meter Subscription – ₹X/month/meter from DISCOMs
- 📊 Utility Intelligence Dashboard – Premium SaaS analytics & insights
- 💸 Fraud-Recovery Linked Model – % share of recovered theft savings
- 🔌 API Licensing to OEMs – Licensing technology to smart meter manufacturers
- 🔄 Cross-Utility Expansion – Scalable to water, gas, and solar infrastructure

MILESTONE ACHIEVED

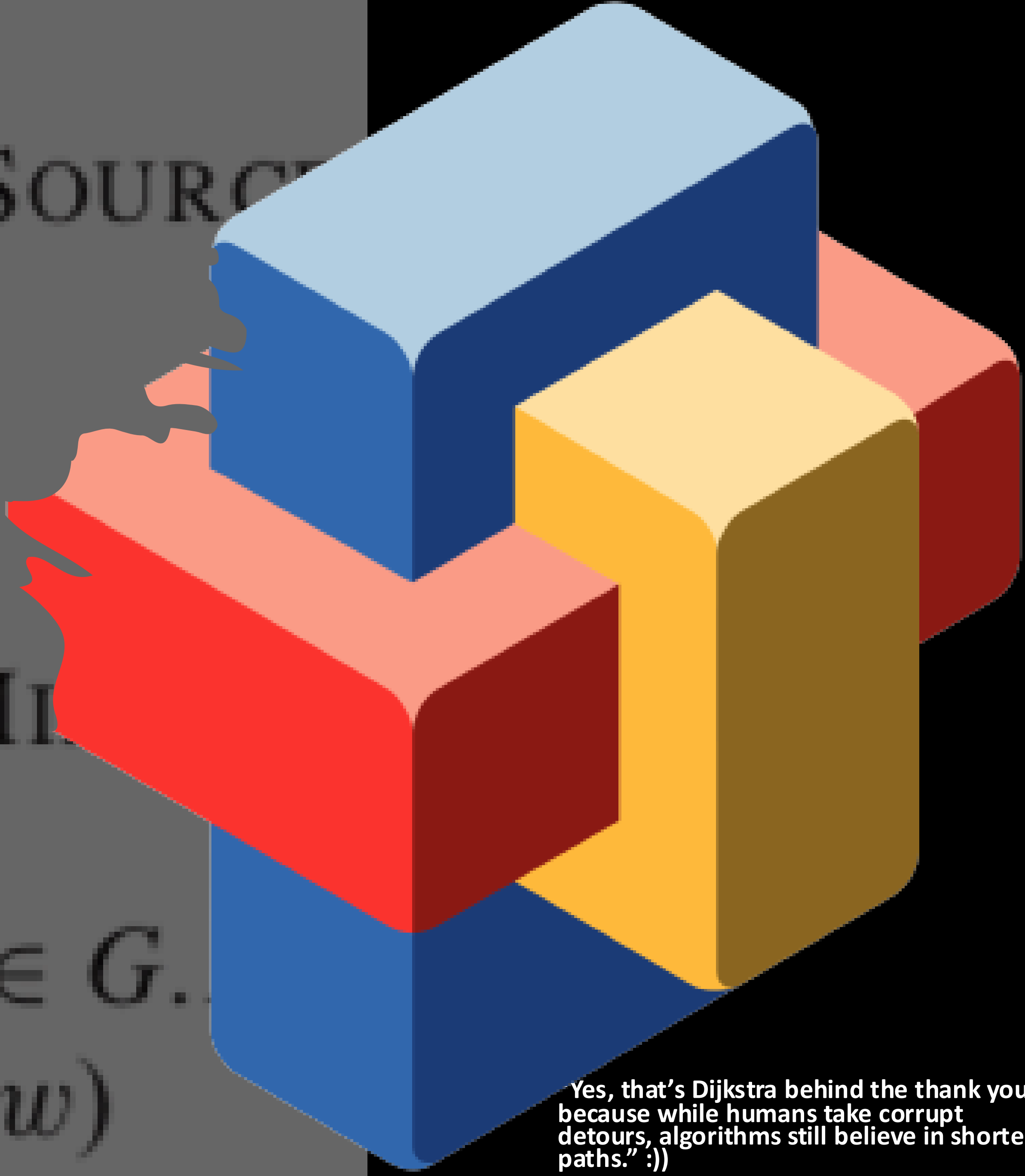
BACKED BY RESEARCH PAPER

- ACHIEVED 96.7 % ACCURACY WITH TESTING ON SIMULATED DATABASE
- **IDEA RECOGNISED AT IRIS NATIONAL FAIR 2024**





THANK YOU



"Yes, that's Dijkstra behind the thank you- because while humans take corrupt detours, algorithms still believe in shortest paths." :))